SMITH–TODA $V(4)$ EXISTS IF AND ONLY IF $p \geq 11$ (CONDITIONAL ON THE $p = 1009$ MANUSCRIPT)

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Abstract. OpenAI's preprint *A finite Smith–Toda complex $V(4)$ at the prime* 1009 chooses $p = 1009$ to make its finite inequalities generous. We check that every finite claim in that proof holds, as written with 1009 replaced by p, for all primes $p \geq 19$. At $p = 11, 13, 17$ three inequalities fail, but the statements they were used for can be verified directly by exact enumeration of May E_1 pages. **Conditional on the correctness of the original $p = 1009$ manuscript, which we did not re-verify**, $V(4)$ therefore exists at every prime $p \geq 11$. Combined with the published nonexistence theorems of Toda, Ravenel and Nave, and a short Adem-relation lemma at $p = 2$, this gives the sharp classification: $V(4)$ exists if and only if $p \geq 11$. Only the "if" direction depends on the manuscript. The argument for $p \geq 11$ is OpenAI's; we only re-select the prime and replace three inequalities by exact enumeration, and we list every changed line. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

1. Statement

Let [1] denote `preprints/A-finite-Smith-Toda-complex-V4-at-the-prime-1009-September-23-2026` of `https://github.com/openai/math` (commit `08fcd61`; line numbers refer to `build/sections/*.tex`) Its main theorem: at $p = 1009$ there is a finite p-local spectrum X with $BP_*X \cong BP_*/(p, v_1, v_2, v_3, v_4)$ as BP_*BP-comodules.

Theorem 1. *Assume that the proof in* [1] *is correct at $p = 1009$. Then for every prime $p \geq 11$ there is a finite p-local spectrum X with $BP_*X \cong BP_*/(p, v_1, v_2, v_3, v_4)$ as graded BP_*BP-comodules, with the canonical quotient coaction and generator in degree zero.*

2. Where p enters

The prime enters [1] only through finite statements about the May E_1 page $\Lambda(h_{i,j}) \otimes \mathbb{F}_p[b_{i,j}]$, with $h_{i,j}$ in bidegree $(1, p^j u_i)$, $b_{i,j}$ in $(2, p^{j+1} u_i)$ and $u_i = (p^i - 1)/(p - 1)$, together with explicit polynomial and divisibility inequalities used to prove those statements. We extracted every such inequality (33 in total; `ineq308.py` lists them with line numbers) and determined, for each, the primes at which it holds. All hold for $p \geq 19$. Exactly three fail somewhere in $\{11, 13, 17\}$:

line	inequality	holds for
`02` L109–113	$p^2 - \frac{27}{2}p - 9 > 0$	$p \geq 17$
`02` L154–155	no nonzero multiple of p in $[-14, 3]$	$p \geq 17$
`03` L161–163	$2p - 19 > p - 2$	$p \geq 19$

The first two are used only to prove Lemma `low:algebraic-vanishing`: certain cobar groups vanish because their May E_1 page is empty. The third is used only in the exhaustion proof of Table `may:n5-table`. At $p = 11, 13, 17$ we verify these conclusions directly. We enumerate every E_1 monomial in every relevant bidegree: $(qk + 1, u_n + k)$ for $n \leq 3$, and $(qk + 1 -$

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$\rho, U+k)$ for every profile $e \in \{0,1,2\}^{n+1}$ with $\rho \geq 2$, for all k up to the normalized cutoff of `low:normalized-cutoff`; also the bidegrees $(q+1, u_4+1)$ and $(q+1, u_5+1)$. The vanishing groups have no monomials. The table at $(q+1, u_5+1)$ has exactly the five entries of [1], with exponents $p-2, p-3, p-3, p-3, p-4$ in place of $1007, \ldots, 1005$, and d_1 is injective over $\mathbb{F}_p$ on the exterior-length-3 and length-5 spans. In the same runs we re-verify the other finite tables the proof uses at each p:

- the v_4-target lists, non-empty exactly at $k = 1, 2, p-1, p$, with the paper's monomials;
- the Frobenius quotient-group table;
- the small-stem lemma, with d_1-cohomology 0 at $(6, W+1)$;
- the page bound $qp + 1 - S = 2p - 6 > 0$, $N - K = p - 3 > 0$.

All other uses of 1009 in [1] are numerical instantiations of these symbolic inequalities (the README lists them line by line). The adversarial review read every section, including the appendix, and found no further dependence on the size of p, in particular no argument of the form "p is large, hence a group vanishes".

Remark 2 (Controls). At $p = 7$ the same enumeration finds non-empty product-obstruction pages, for example $y\,A^{16}$ in bidegree $(33, 119)$ for the profile $e = (0, 2, 0, 2)$, $k = 3$. So the mechanism fails at $p = 7$. This is consistent with Toda's requirement $p \geq 11$ for a unital product on $V(3)$ (as reported in [4, §5.6, p. 184]) and with Nave's theorem that $V((p+1)/2) = V(4)$ does not exist at $p = 7$. At $p = 1009$ the enumeration reproduces the paper's tables verbatim (positive control).

Remark 3 (What is not claimed). We do not verify the homotopy-theoretic content of [1]: the adjusted-filtration injectivity (§3), the Frobenius left-hand side (§4), the extension of powers (§5) and the appendix pairing. Theorem 1 is exactly as reliable as that proof. The primes $p \leq 7$ are treated in §3.

3. The classification

We use the standard definition [2, Definition 1.1]: a Smith–Toda complex $V(k)$ is a finite p-local spectrum with $BP_*V(k) \cong BP_*/(p, v_1, \ldots, v_k)$ as a BP_*-module (hence as a BP_*BP-comodule). This is the notion in Theorem 1. The definition does not refer to $V(k-1)$.

Lemma 4 ([2, Theorem 1.2 and p. 493]). Let p be odd. If $V(k)$ exists, then it is the cofiber of a map $\Sigma^{|v_k|}V(k-1)' \to V(k-1)$ inducing v_k on BP-homology, where $V(k-1)$ and $V(k-1)'$ are Smith–Toda complexes (not necessarily equivalent). In particular, if no $V(k-1)$ exists at p, then no $V(k)$ exists at p.

Nave proves that $BP_*X \cong BP_*/I_{k+1}$ if and only if $H_*(X; \mathbb{F}_p) \cong E(\tau_0, \ldots, \tau_k)$ as an A_*-comodule. The bottom part of a minimal cell structure, the cells below dimension $|\tau_k| = 2p^k - 1$, then has homology $E(\tau_0, \ldots, \tau_{k-1})$, because every monomial without τ_k has degree at most $\sum_{i<k}(2p^i - 1) < 2p^k - 1$. So that skeleton is a $V(k-1)$.

Lemma 5. Let $p = 2$ and $k \geq 1$. There is no finite 2-local spectrum X with $BP_*X \cong BP_*/(2, v_1, \ldots, v_k)$ as a BP_*-module.

Proof. We use only $BP_iX = 0$ for $i < 0$ and $1 \leq i \leq 3$ (here $k \geq 1$ is used for $i = 2$), and $BP_0X = \mathbb{Z}/2$. This is Ravenel's Adem-relation argument for $V(1)$ [4, §1.3, p. 19], written out so that it visibly applies for every k.

(i) The cofiber C of $BP \to H\mathbb{Z}_{(2)}$ is 2-connected. If m is minimal with $H_m(X; \mathbb{Z}_{(2)}) \neq 0$, then $BP_mX \cong H_mX$, so $m = 0$, and X is (-1)-connected. Since $C \wedge X$ is 2-connected, $BP_iX \to H_i(X; \mathbb{Z}_{(2)})$ is an isomorphism for $i \leq 1$ and onto for $i = 2$. Hence $H_0 = \mathbb{Z}/2$ and $H_1 = H_2 = 0$ integrally, and $H_*(X; \mathbb{F}_2)$ is $\mathbb{F}_2$ in degrees 0 and 1 and 0 in degree 2.

(ii) Take a minimal cell structure: one cell per basis element of $H_*(X;\mathbb{F}_2)$, so all mod-2 cellular differentials vanish. Then $X^{(2)} = M(2)$, $Sq^1 x_0 = x_1$, and $X^{(3)} = M(2) \cup_{g_j} \bigvee_j e_j^3$ with $g_j \in \pi_2 M(2) \cong \mathbb{Z}/4\langle\tilde{\eta}\rangle$. Here $q\tilde{\eta} = \eta$ for the pinch map $q\colon M(2) \to S^1$, and $2\tilde{\eta} = i\eta^2$. The BP-Hurewicz image of $i\eta^2$ in $BP_2M(2) = \mathbb{Z}/2\langle v_1\rangle$ is 0. Since $BP_3(X/X^{(3)}) = 0$, the map $BP_2X^{(3)} \to BP_2X = 0$ is injective, and $BP_2X^{(3)} = BP_2M(2)/\langle h(g_j)\rangle$. So $h(g_j) = v_1$ for j in a non-empty set J. For $j \in J$, g_j is an odd multiple of $\tilde{\eta}$ and $qg_j = \eta$. Since Sq^2 detects η on the subquotient $X^{(3)}/X^{(0)}$, we get $y := Sq^2 x_1 = \sum_{j\in J} e_j^* \neq 0$.

(iii) $BP_3X^{(3)} \cong K := \{\sum a_j e_j \in \bigoplus_j \mathbb{Z}_{(2)} e_j : \sum_{j\in J} a_j \equiv 0\ (2)\}$, and $BP_3X^{(4)} \hookrightarrow BP_3X = 0$. Hence the integral cellular boundary $d\colon C_4 \to C_3$ has image $K \ni 2e_{j_0}$ ($j_0 \in J$). Write $\tilde{y} = \sum_{j\in J} e_j^*$ and pick c with $dc = 2e_{j_0}$. Then $(\delta\tilde{y}/2)(c) = 1$, so $Sq^1 y = [\delta\tilde{y}/2] \neq 0$.

(iv) The Adem relations give $Sq^3Sq^1x_0 = Sq^3x_1 = Sq^1Sq^2x_1 = Sq^1y \neq 0$. But also $Sq^3Sq^1x_0 = Sq^2Sq^2x_0 \in Sq^2H^2(X;\mathbb{F}_2) = 0$. This is a contradiction. □

Corollary 6. Assume the proof in [1] is correct. Then a Smith–Toda complex $V(4)$ exists at the prime p if and only if $p \geq 11$. The "only if" direction is unconditional.

p	$V(4)$	reason
2	no	Lemma 5 (no $V(k)$ for any $k \geq 1$; $V(1)$: Toda [6])
3	no	Lemma 4 twice; $V(2)$ does not exist at $p = 3$ (Toda [6]; see [2, p. 493], [4, §1.3])
5	no	Lemma 4; $V(3)$ does not exist at $p = 5$ [4, Theorem 7.6.1]
7	no	[2, Theorem 1.3]: $V((p+1)/2) = V(4)$ does not exist for $p \geq 7$
≥ 11	yes	Theorem 1 (conditional on [1])

At $p = 7$ the Annals version is needed. The 1998 preprint [3] only proves that $V((p+3)/2)$ does not exist for $p > 5$ (Theorem 1.1 there), and at $p = 7$ that is $V(5)$. Mahowald reportedly conjectured that no $V(n)$ with $n > 3$ exists at any prime [5]. The "if" direction contradicts that conjecture, which is another reason the homotopy theory of [1] needs independent verification.

4. Certificates

`may308.py` (direct recursive enumeration, exact integers, d_1 ranks over $\mathbb{F}_p$) and `may308_dp.py` (knapsack DP counts) form the primary computation. `independent/sweep.py` and `independent/d1check.py` were written separately; they use z3 exact enumeration with blocking clauses and an int64 DP modulo $2^{61} - 1$. All agree at $p = 7, 11, 13, 17$ and at $p = 1009$ (sampled). `may308.py` also passes at $p = 19, \ldots, 47$.

References

[1] OpenAI, *A finite Smith–Toda complex $V(4)$ at the prime* 1009, OpenAI Math Release preprint, September 23, 2026, `https://github.com/openai/math/tree/main/preprints/A-finite-Smith-Toda-complex-V4-at-the-prime-1009-September-23-2026`.

[2] L. S. Nave, *The Smith–Toda complex $V((p+1)/2)$ does not exist*, Ann. of Math. 171 (2010), 491–509.

[3] L. S. Nave, *On the nonexistence of Smith–Toda complexes*, arXiv:math/9810178 (1998).

[4] D. C. Ravenel, *Complex cobordism and stable homotopy groups of spheres*, 2nd ed., AMS Chelsea, 2004.

[5] A. Salch, *The cohomology of the height four Morava stabilizer group at large primes*, arXiv:1607.01108 (2016).

[6] H. Toda, *On spectra realizing exterior parts of the Steenrod algebra*, Topology 10 (1971), 53–65.